

Redefining Cognition as an Interactive Feedback System

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Abstract

Traditional neuroscience has long viewed brain neuronal networks as the sole substrate for cognition. While recent theories of Basal Cognition challenge this encephalocentric view by demonstrating cognitive-like behaviors in brainless organisms (e.g., slime molds, planarians), they fall into a fatal logical dilemma regarding non-living physical materials that exhibit identical adaptive physics (e.g., shape-memory alloys). This paper resolves this contradiction by redefining **cognition** not as a biological privilege, but as a universal **Interactive Feedback Process** that maintains system identity against external and internal stimuli. We introduce a quantitative model, $T = C \times \Delta t$, proving that the difference between non-living matter and complex biological organisms is a quantitative spectrum of complexity (C) and time horizon (Δt), rather than a binary presence of mind. Furthermore, we reframe biological phenomena—such as sleep, immune response, and forgetting—as mechanical energy-rerouting and optimization feedback algorithms.

1. Introduction

Unlike early ancestors who hunted with sharpened sticks and stones, modern humans communicate across continents via palm-sized smartphones, immerse themselves in virtual reality, and interact with conversational AI. Humanity has erected a remarkably sophisticated civilization, expanding at an accelerating pace.

Amidst this evolutionary progression, a fundamental question arose:

"Why did only humans achieve such a highly developed civilization?"

Historically, science concluded that human progress was driven by highly evolved cognition centered strictly in the brain, where complex neuronal interactions generate thought. This brain-centric paradigm was accepted as an undisputed fact for decades.

Recently, the emerging field of **Basal Cognition** shattered this neurocentric boundary. By demonstrating that brainless organisms such as **slime molds** and single-celled amoebas can perceive stimuli, make decisions, and adapt, **research in basal cognition** proved that cognitive phenomena occur at the cellular level.

However, the logic of Basal Cognition contains a fatal vulnerability. This paper addresses this flaw and establishes a unified, comprehensive definition of cognition spanning both biological and physical systems.

2. Main Body

2.1 The Logical Bankruptcy of Basal Cognition

Basal Cognition argues that single-celled organisms—such as *Physarum polycephalum* (slime mold), *Stentor*, and amoebas—possess cognition because they autonomously perceive stimuli, judge environments, and adapt.

However, all biological entities are fundamentally composed of non-living matter that transformed its structural and chemical organization. Before claiming that cognition is an exclusive property of biological cells, one must examine non-living physical materials.

Inanimate objects also alter or maintain their structure upon receiving

stimuli. Shape-memory alloys, memory gels, and stimulus-responsive polymers absorb external inputs (heat, pressure, electrical charge) to restore their original forms. Principle-wise, this response is identical to cellular feedback.

A conventional counterargument claims: *"Physical responses in materials are merely laws of physics, not cognition."*

However, this counterargument collapses when applied to single-celled slime molds (*Physarum polycephalum*). Slime mold navigation and decision-making are driven entirely by fluid dynamics: intracellular fluid oscillations, pressure gradients, and ion diffusion—pure physical laws.

This traps Basal Cognition in an inescapable logical checkmate:

1. If cellular physical responses are accepted as cognition, then non-living materials (e.g., shape-memory alloys) reacting via identical physical laws must also be recognized as cognitive. This destroys the premise that *"only biological cells possess cognition."*

2. If material responses are dismissed as mere physics, then slime mold responses—driven by the same fluid dynamics—must also be classified as *"mere physical phenomena,"* destroying the primary evidence supporting Basal Cognition.

Regardless of which path is taken, Basal Cognition invalidates its own premises.

2.2 A New Definition of Cognition and Quantitative Mathematical Model

To resolve this limitation, we propose a universal definition of cognition that transcends cellular boundaries to encompass the physical universe:

Cognition:

The overall **Interactive Feedback Process** in which a system maintains its identity and reconstructs its structure against external and internal stimuli.

Cognition is not an exclusive property of cells, but the interactive process of a system maintaining itself against stimuli. To quantify the difference between non-living materials and human thought, we introduce the **Total Cognition Formula**:

$$T = C \times \Delta t$$

T: Total Cognition (The total magnitude of interactive feedback exerted by the system)

C: Complexity (The complexity of interactive feedback circuits formed between internal particles or cells)

Δt : Time Horizon (The spatiotemporal range covered, from absorbing/remembering past stimuli to predicting/responding to future stimuli)

Applying relative values demonstrates that the distinction between inanimate objects and humans is a numeric spectrum rather than a binary presence of mind:

Passive Material (Shape-Memory Alloy):

Simple internal circuit ($C = 10$), short-term reaction immediately following stimulus ($\Delta t = 0.1$).

$$T = 10 \times 0.1 = 1$$

Advanced Biological Organism (Human):

Complex network of tens of trillions of cells and neurons ($C = 1,000,000$), predicting decades into the future based on past memory ($\Delta t = 100$).

$$T = 1,000,000 \times 100 = 100,000,000$$

Thus, non-living matter and biological life are divided not by the presence or absence of thought, but by quantitative differences in computational scale (T) generated by complexity (C) and time horizon (Δt).

2.3 System Spectrums and Physical Mechanisms of Biological Phenomena

Interactive systems are classified into two spectrums based on stimulus generation:

Level 1 System (Passive / Inanimate Material): Activates feedback only upon incoming external stimuli → Small T value.

Level 2 System (Active / Biological Life): Generates internal stimuli via metabolism autonomously → Massive T value.

Biological phenomena previously treated as simple life traits are cleanly

explained via specific physical mechanisms:

1. Optimization Feedback Mechanism of Sleep and Unconsciousness

Sleep and fainting are not the cessation of thought. They represent a system optimization mechanism that shuts down external input circuits and reroutes cognitive energy toward internal repair circuits to restore system damage.

2. Energy Rerouting Mechanism of the Immune Response

Lethargy and reduced higher-level cognitive performance during infection are not a loss of intelligence. They are the physical result of rerouting computational energy away from higher cognitive circuits toward immune feedback circuits to prevent system destruction by pathogens.

3. Noise Filtering Mechanism of Forgetting

Permanently storing all stimuli causes internal computational overload. Forgetting is not information loss, but a physical system optimization algorithm that deletes unnecessary noise to maximize computation speed for key feedback peaks.

3. Conclusion

This study exposes the logical checkmate of Basal Cognition and redefines cognition as a physical interactive feedback system:

1. Cognition is not an exclusive privilege of brains or cells, but the

feedback process of maintaining system identity against stimuli.

2. The magnitude of cognition is quantified by $T = C \times \Delta t$, demonstrating that differences between materials and humans are numerical spectrums (T).

3. Sleep, immune responses, and forgetting are precise physical feedback algorithms designed for system sustainability.

This framework establishes a foundation unifying biology and physics.

Building upon this model, subsequent papers will explore:

Part 2: System compatibility in organ transplantation, time-freezing in cryonics, and coexistence with AI equipped with self-circulating circuits.

Part 3: The expansion of this framework into the **Cosmic Cognition Hypothesis**.

References & Data Sources

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